

Email: shangwen.sun@nyu.edu Homepage: savinusun.github.io

Education

Baruch College, City University of New York New York, NY
Master of Science in Financial Engineering 08/2021 – 12/2022
 Selected Courses: Machine Learning, Natural Language Processing, Stochastic Processes

School of Mathematical Sciences, Peking University Beijing, China
Bachelor of Science in Mathematics and Applied Mathematics 09/2016 – 07/2021
Bachelor of Economics Double Degree
 Selected Courses: Differential Equations, Advanced Algebra, Bayesian Theory

Work Experience

New York University New York, NY
Research Assistant 04/2025 – Present
Hosted by Prof. Alfredo Canziani and Prof. Yann LeCun

- Conducted exploratory experiments on open-source LLM models, investigating mechanisms underlying massive activations and attention sinks.
- Wrote PyTorch code for multi-node LLM training using FSDP to run ablation studies on Llama-style architectures.

Millennium Management New York, NY
Quantitative Researcher, Model-Driven Algorithmic Trading Team 01/2023 – 03/2025
Quantitative Developer Intern, Statistical Arbitrage Platform 06/2022 – 08/2022

- Conducted research on U.S. equity order execution and built batch-parallelized research tools, enabling faster tick-level experiments and more efficient strategy evaluation.
- Designed and deployed short-horizon signals for an in-house VWAP algorithm, reducing execution costs on targeted order flows, and built post-trade analysis models that incorporated historical flow features to assess broker performance.
- Performed P&L attribution and seasonal portfolio analysis across customized risk factors and volatility regimes, and corrected errors in an in-house Barra risk model.

Publications

Anatomy of Massive Activations and Attention Sinks *Accepted ICML 2026*
Shangwen Sun, Alfredo Canziani, Yann LeCun, Jiachen Zhu
 We provide a mechanistic account of why massive activations and attention sinks co-emerge in pre-trained LLMs. We show that normalization links the two by converting massive activations into near-constant hidden states that can induce sinks. We find that attention-space geometry and training context lengths shape sink behavior, biasing some heads toward short-range structure. Since the two phenomena can be reduced independently without hurting modeling performance, their overlap appears incidental rather than functionally necessary.

The Case for Locality: Not All Attention Should Be Global *Under Review*
Shangwen Sun, Alfredo Canziani, Yann LeCun, Jiachen Zhu
 We propose a hybrid attention block that replaces a subset of global attention heads with small-window sliding attention and convolutional heads. A conditional gating mechanism dynamically balances the contributions of different head types based on the input. We show that using only local attention or only gated attention gives only marginal gains, while combining them makes locality explicit, trainable, and computationally efficient.

Selected Awards

Meritorious Winner, The Mathematical Contest in Modeling
 Gold Medal, 14th Chinese Girls Mathematical Olympiad
 Silver Medal, 31st Chinese Mathematical Olympiad

Activities

Vice President, European Society, Peking University 2017 – 2018